

Assets, Threats and Risk

What matters? What can harm it? How serious is it?

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Three questions organize today's lecture

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1

Assets

What is valuable? Identify four asset types, then explain their value and needs.

2

Harm / controls

How could harm occur? Learn four roles, then examine protection.

3

Risk

How likely and how serious? Define two ratings, then combine them.

One hospital story connects the lesson

WannaCry disrupted NHS services in May 2017.

NHS = National Health Service, England's public healthcare system.



1 ·Assets

Identify assets → classify them → explain their protection needs.

Asset = something valuable to protect

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Something valuable that needs protection.

NHS example: records, computers, staff and healthcare services all support care. We will group them into four asset types.

Four asset types: what are we protecting?

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Assets take different forms. These four categories help us identify what needs protection before choosing protection.

Information: useful facts we depend on

Examples: patient records, grades, passwords and appointment details

Real Case: Equifax 2017

Devices: physical tools that support work

Examples: laptops, phones and hospital computers

Real Case: NHS services in May 2017

People: knowledge, work and wellbeing

Examples: People provide knowledge, skills and essential work

Real Case: NHS services in May 2017

Services: functions people rely on

Examples: online exams, messaging, payments and appointments

Real Case: On 4 October 2021, Facebook services became unavailable

Why an asset list is only the beginning

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1	**Value**	Why does the asset matter? NHS systems support care.
2	**Dependency**	What does it need to work? A service needs working devices and networks.
3	**Protection goals**	What must be preserved? Use confidentiality, integrity and availability.

Value, dependency, and protection

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Value

- Imagine two similar computers: one shows a **public notice,** and one supports an **urgent service**.
- Their prices may be the same, but interruption has different consequences

Dependency

- A dependency is something else you need.
- Facebook's October 2021 outage showed why these connections matter.
- It is possible to have several services that all rely on the same underlying system. They can fail together

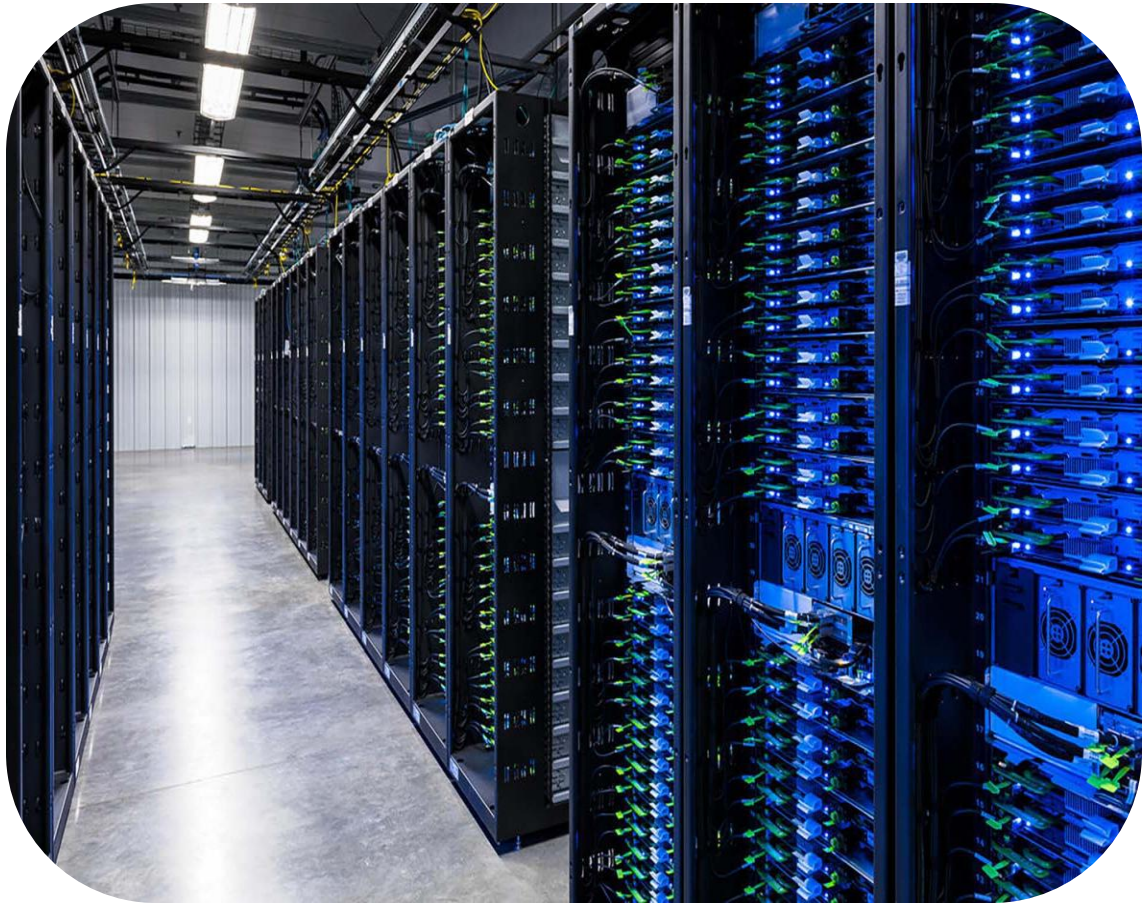
Facebook: devices alone were not enough

Data center = a building with computing equipment.

Facebook, 4 October 2021: a network setting change interrupted communication between data centers.

Services became unavailable.

Dependency: services need communication.



Week 1 link: what can an asset lose?

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C	**Confidentiality**	Only permitted people can access it. **Equifax:** private data was exposed.
I	**Integrity**	Information stays correct and complete. **GitLab:** database data was deleted.
A	**Availability**	It is usable when needed. **NHS:** healthcare work was disrupted.

Activity: identify the assets

Real case: WannaCry disrupted NHS healthcare in 2017.
Use the four asset categories to explain what needed protection.

1. Name one information asset.
2. Name one device asset.
3. Name the people affected.
4. Name the service and its value.

Connect each asset to its value

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1. **Information**: patient and appointment records support decisions.
2. **Devices**: computer systems let staff use that information.
3. **People:** staff provide knowledge and care; patients are affected.
4. **Services**: healthcare must remain usable when needed.

Next: what could harm these assets?

We can name what matters. Now explain what could harm it.

Known: assets and their value.

Next: threats, weaknesses and exploits.

Then: controls that reduce the harm.

2 ·Harm and Protection

Explain harm → identify weaknesses → examine controls.

Four concepts explain harm and protection

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We know what is valuable. Now ask how it could be harmed.
These four roles explain the Equifax incident step by step.

- **Threat:** a possible cause of harm
- **Vulnerability:** a weakness
- **Exploit:** a way to use a weakness
- **Control:** a protective measure

Threat = a possible cause of harm

Something that could cause harm to an asset.

Threats may be deliberate, accidental or physical. We will compare WannaCry, GitLab and a data-center fire.

Threats can be deliberate or accidental

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M

Malicious

Deliberate harm.
WannaCry, 2017: harmful software disrupted services.

A

Accidental

Unintended harm.
GitLab, 2017: database data was deleted by mistake.

P

Physical

Physical damage.
Kakao, 2022: a data-center fire disrupted services.

A harmful software

A mistake causes harm

Buildings and power matter

WannaCry: malware and ransomware

Malware = software designed to cause harm.

Ransomware = malware that blocks access and demands payment.

WannaCry disrupted NHS work in May 2017.



Real WannaCry screen • Microsoft, 2017

Vulnerability = a weakness that permits harm

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A weakness that could allow harm.

Equifax, 2017: an unfixed software flaw allowed attackers to gain access.

Threat and vulnerability are different

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Threat

What could cause harm?

Equifax, 2017: attackers sought access to personal information.

Vulnerability

What weakness could permit harm?

Equifax's affected web software had a flaw that had not been fixed.

Attackers are the threat and a flaw in the software is the weakness attackers used.

Exploit = take advantage of a weakness

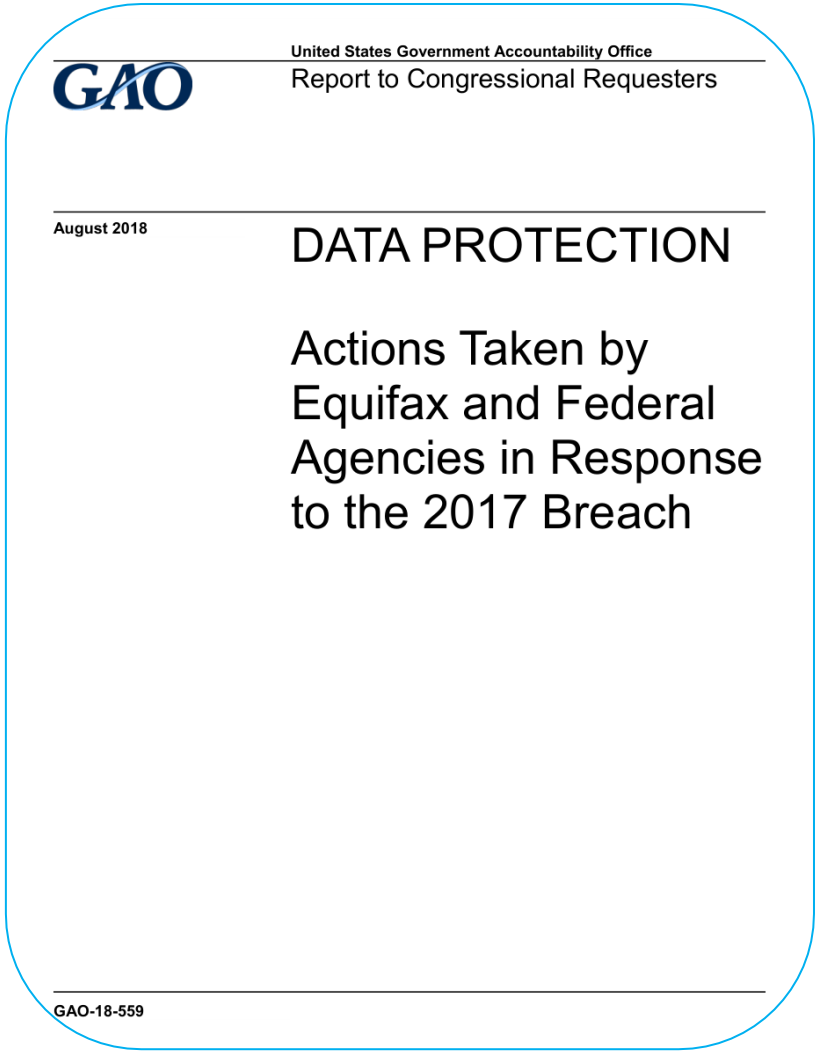
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An exploit is a method or tool that uses a vulnerability.

To exploit a flaw means to use it to cause an unintended result.

Equifax: attackers used the flaw to gain access.

Vulnerability:	**the flaw**
Exploit:	**its use**



United States Government Accountability Office

GAO

Report to Congressional Requesters

August 2018

DATA PROTECTION

Actions Taken by Equifax and Federal Agencies in Response to the 2017 Breach

GAO-18-559

When possible harm becomes an incident

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Incident

Something has happened that harms or threatens security.

WannaCry disrupted NHS services in 2017.

Possible future risk

Possible future harm, judged by its chance and seriousness.

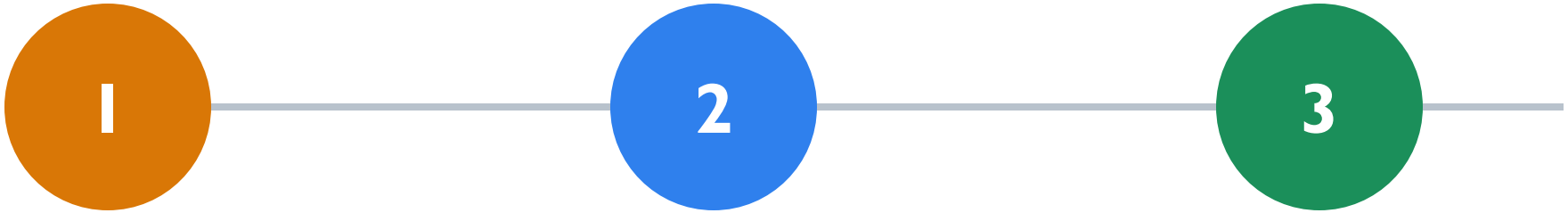
A similar exposed system might suffer disruption.

Control = a measure that reduces harm

- Protection that lowers the chance or seriousness of harm.
- A software fix can remove a known weakness.
- Control makes exploitation through that weakness less likely.

Controls can prevent, detect or recover

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Stop harm before it occurs.
Example: fix a known software flaw.

Notice a problem.
Example: check records of unusual access.

Restore useful work.
Example: recover data from a tested copy.

Patch = an update that fixes a problem

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Patch and patching

A patch is a software update that fixes a problem.

Patching means applying the update.

Equifax lesson

A patch instruction was not enough.

The affected system needed the fix, and completion needed checking.

Why account access needs its own controls

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Patching protects software. Accounts also need protection.
These terms explain the Uber and Cloudflare login cases.

Phishing: deceptive messages

Credentials: identity evidence

Authentication: checking identity

MFA: different kinds of evidence

Phishing and credentials: two useful words

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Phishing

A deceptive message or website tries to trick someone.

It may ask for a password or a login approval.

Credentials

Evidence used to prove identity, such as login details.

Cloudflare's 2022 attackers tried to steal staff credentials.

Authentication = checking who is signing in

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1

Authentication

The process of checking a claimed identity

2

MFA

Multi-factor authentication: more than one kind of identity evidence.

3

Example

Password = what you know.
Security key = a device you have.

Uber: an unwanted login was approved

Contractor = an outside worker.
Compromised = accessed without permission.

- Uber, September 2022: after repeated requests, a contractor approved a login. An attacker gained account access.



Check unexpected login requests.

Cloudflare: a key checked the real website

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Security key = a physical login device.

Cloudflare, 2022: staff submitted login details, but keys blocked the fake-site login.



Origin-bound = works only where it belongs

Compare what the control actually requires

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Cloudflare • 2022

The security key checked the website's identity.

Stolen login details were insufficient for that route.

Uber • 2022

After repeated requests, a person approved a login.

The attacker gained access to an account.

Why recovery follows prevention

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Prevention can fail. Recovery limits the harm.

GitLab's deletion raises a new question: can useful data be recovered? We need two terms: backup and restore.

Backup and restore are different actions

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Backup

Make a separate copy for recovery.

A completed copy is only the beginning of a recovery plan.

Restore

Use a copy to recover usable data or service.

GitLab's 2017 incident showed why recovery must be tested.

Practice: label the Equifax incident

Equifax, 2017: attackers used an unfixed software flaw to gain access to personal information.

1. Asset: what was valuable?
2. Threat: who or what could harm it?
3. Separate weakness from exploit.
4. Control: what would address the flaw?

Answers: each term has a different job

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1. **Asset:** personal information that needed protection.
2. **Threat source:** attackers seeking access.
3. **Vulnerability:** the unfixed software flaw; **exploit:** using it.
4. **Control:** a verified patch addresses that flaw.

3. Assessing Risk

Judge possible harm: how likely and how serious?

Three tools for a simple risk assessment

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1	**Likelihood**	The chance of a harmful event. **Ask:** how likely is it?
2	**Impact**	The seriousness of the consequences. **Ask:** how much harm?
3	**Risk matrix**	A grid that combines both ratings. Use it to compare attention needed.

Define the event before rating it

1	**Event**	What harmful event do we mean? **Example:** an attacker gains account access.
2	**Period**	Over what period? For example, before next month's exam.
3	**Current controls**	What protection is working now? Cloudflare's keys blocked one login route.

Separate facts, assumptions and unknowns

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F

Fact

Supported by evidence.
Equifax's affected software flaw was not fixed.

A

Assumption

Accepted for this exercise.
A similar system has a reachable weakness.

?

Unknown

Missing evidence.
Has the relevant fix been applied and checked?

Likelihood: use three clear labels

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1

Unlikely

The event is not expected under the stated conditions and controls.

2

Possible

The event could happen; a believable route to harm exists.

3

Likely

The event is expected or strongly supported by evidence.

Impact: use three clear labels

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1

Minor

Small disruption; routine recovery; little lasting harm.

2

Serious

Important work or privacy is affected; extra help or time is needed.

3

Severe

Essential service loss or major harm that is difficult to reverse.

Real incidents show different kinds of impact

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C

Care

NHS, 2017:
healthcare disruption affected appointments and care.

Can care continue?

P

Privacy

Equifax, 2017:
personal information was exposed.

Who can see private data?

W

Work

GitLab, 2017:
lost database changes affected users' work.

What work must be redone?

Risk matrix: combine two ratings

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Likelihood ↓ Impact →	1• Minor	2• Serious	3• Severe
3• Likely	3• Medium	6• High	9• High
2• Possible	2• Low	4• Medium	6• High
1• Unlikely	1• Low	2• Low	3• Medium

Worked risk example: hospital disruption

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1	**Likelihood: 3**	Assume an active threat can reach a system missing the relevant fix.
2	**Impact: 3**	Assume an essential care service would stop and recovery would be slow.
3	**Score: 9**	3 × 3 = 9 → High.

Assumed ratings: likelihood (L), impact (I)

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1. WannaCry-like disruption: essential care stops. **L3 I3**

2. GitLab-like deletion: a recent restore works. **L2 I1**

3. Cloudflare-like route: strong key protection. **L1 I3**

4. Update failure: essential work is interrupted. **L2 I3**

A patch can reduce likelihood

After verification

The relevant patch is applied and checked.

That exploitation route is reduced; other weaknesses may remain.

Before the control

Equifax.

A reachable unfixed flaw creates an attack opportunity.

A usable restore can reduce impact

With tested recovery

Classroom example: a recent protected copy restores successfully.

Less work may be lost, but some harm can remain.

Without usable recovery

GitLab's 2017 incident reminds us that lost recent data can remain lost.

Recovery gaps can increase harm.

Review: Six terms you should be able to explain

Asset = something valuable.

Threat = possible cause of harm.

Vulnerability = a weakness.

Exploit = a way to use a flaw.

Control = a protective measure.

Risk = possible harm judged by likelihood and impact.

Practice: explain before you calculate

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1. Why are a computer and its stored records different assets?
2. In Equifax, what was the vulnerability and what was exploitation?
3. Does a tested restore mainly reduce likelihood or impact?
4. What does likelihood 2 × impact 3 mean in our matrix?

Answers: connect the word to the meaning

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1. A computer is a device; its records are information.
2. The flaw was the vulnerability; using it was exploitation.
3. A usable restore mainly reduces impact: less harm after loss.
4. 2 × 3 = 6 → High: possible event, severe consequences.

Assignment: assess an online exam

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Fictional exercise • Exam in 30 days
A public exam system has an unfixed software flaw.
A backup exists, but nobody has tested restoring it.

1. Name the four asset types.
2. Explain a possible harmful event.
3. Rate likelihood and impact.
4. Choose and explain a control.

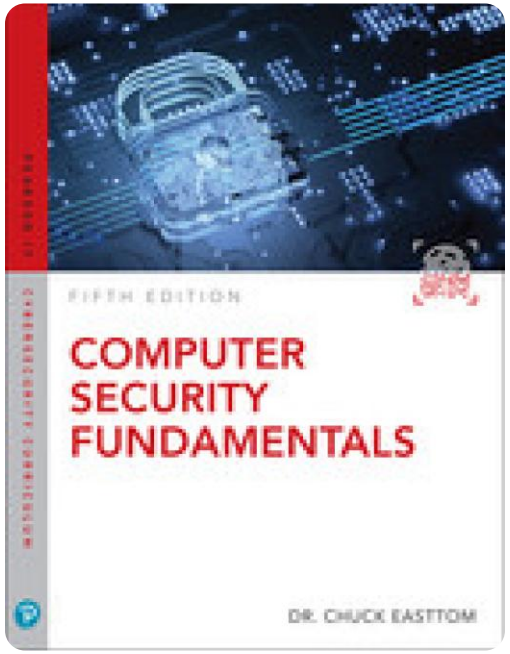
Report your reasoning in four sentences

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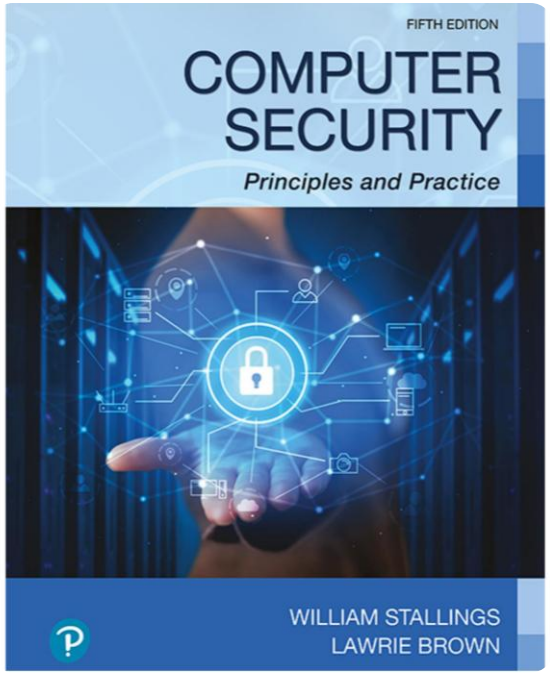
Use the terms, then explain them in plain English.

1. “The asset is …; it matters because …”
2. “The threat could use …, which is the weakness.”
3. “Likelihood is … and impact is … because …”
4. “Our control is …; it reduces …”

Reading: use the books to review the terms



Chuck Easttom
Computer Security
Fundamentals
5th edition



Stallings & Brown
Computer Security:
Principles and Practice
5th edition



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Thank you!